From Clear to Adverse: A Survey of All-Weather LiDAR Semantic Segmentation for Autonomous Driving

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ABSTRACT

This paper presents a comprehensive survey of adverse-weather robustness for 3D LiDAR semantic segmentation in autonomous driving. With the push toward safe, trustworthy, and operationally reliable autonomy beyond clear-day operation, robust 3D perception under fog, rain, snow, spray, or dust has emerged as a core barrier to expanding the operational design domain (ODD) and providing safety assurance in real-world deployment, rather than a purely algorithmic objective. LiDAR segmentation performance can degrade sharply due to structured sensing degradation, including precipitation-induced clutter, range-dependent attenuation, wet-surface ground dropouts, and radiometric distortions in the intensity channel. To support systematic research and fair comparison, we first formulate the task and clarify the most common learning protocols—domain generalization (DG), unsupervised domain adaptation (UDA), and test-time adaptation (TTA)—which differ in when target-domain information is accessible. We then analyze weather-induced domain shift from both a physics/measurement perspective and a learning perspective, and distill practical design implications. Building on this analysis, we organize representative methods with a protocol-aware, two-axis taxonomy that couples the learning setting (DG/UDA/TTA) with the pipeline intervention level (input/preprocess, representation, and objective/prediction). We further summarize datasets and evaluation metrics, and compile reported cross-domain performance on a common real-world adverse-weather benchmark (SemanticSTF) under as consistent a backbone setting as possible. Finally, we close with a benchmarking-oriented synthesis, and open research directions toward reliable all-weather LiDAR semantic segmentation.

1. Introduction

Autonomous driving is moving from curated demonstrations toward safety-assured, large-scale operation in the open world, where the operational design domain (ODD) must extend well beyond clear days and controlled settings. In real traffic, weather can shift rapidly and unpredictably, and adverse conditions—fog, rain, snow, spray, or dust—remain a persistent and costly obstacle to deployment because they fundamentally alter sensing physics and amplify perception uncertainty. Among perception modules, 3D LiDAR semantic segmentation is safety-critical: it provides point-wise scene semantics that shape drivable-space reasoning and object understanding for downstream planning and control. When segmentation fails under adverse weather, the impact is not merely an accuracy drop: it can distort free-space boundaries and corrupt obstacle semantics, propagating errors into decision-making. Improving adverse-weather robustness for LiDAR semantic segmentation is therefore a grand challenge for reliable all-weather autonomy.

Fig. 1 illustrates how adverse weather degrades LiDAR point clouds. In fog, a dense ring of near-range returns surrounds the ego vehicle (see the zoom), likely caused by droplet backscatter that scatters and blocks beams. With many beams disturbed early, fewer valid returns reach the scene, so background structures and object shapes become sparse and incomplete. In snow, clutter near the vehicle remains strong, but it appears as scattered, speckle-like returns along the scan, consistent with intermittent echoes from snowflakes. These examples motivate the framing of this survey. Because the sensing degradation is complex and non-uniform, a careful, mechanism-aware view of weather effects is essential, and a dedicated survey is needed to organize and compare robust segmentation methods under such conditions.

Despite a rich toolbox for 3D robustness—most notably domain generalization (DG) and domain adaptation (DA)—and multiple reviews summarizing these directions, the adverse-weather setting remains insufficiently addressed. Existing methods and surveys are largely developed and validated for clear-weather cross-dataset transfer, where domain gaps mainly arise from scene statistics, sensor setups, or annotation conventions [53, 55, 46, 28]. In adverse weather, the domain shift is qualitatively different: the observations themselves become unreliable, with missing returns, spurious measurements, and strong range-dependent degradation. This mismatch can undermine common DG/DA assumptions. Invariance-driven training may suppress weak but useful cues or overfit to weather artifacts, leading to negative transfer, while self-training and alignment can accumulate errors when pseudo-labels are produced from corrupted inputs, causing boundary drift [29, 39]. These issues make weather-aware design and evaluation essential, including clearer guidance on where to intervene in the pipeline and how conclusions change across learning protocols.

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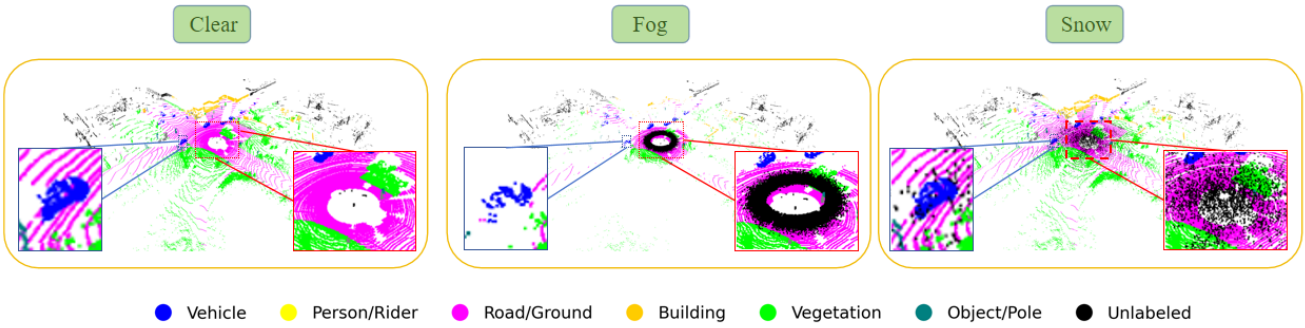


Figure 1: Qualitative visualization of adverse-weather effects on LiDAR point clouds. Compared with clear conditions, fog and snow introduce structured artifacts such as near-range clutter and missing/fragmented returns, altering the spatial evidence available for semantic segmentation.

Current surveys do not yet provide a protocol-aware, segmentation-centric account of adverse-weather robustness for LiDAR perception. Many 3D adaptation/generalization surveys are scoped around broad cross-domain transfer (sim-to-real, cross-sensor, or cross-dataset), where multiple factors are intertwined and discussed under a unified adaptation lens [53]. Weather-centered surveys, on the other hand, provide valuable overviews of adverse-condition sensing and applications, but they often remain at a general level and do not connect weather-induced measurement mechanisms to segmentation-oriented learning protocols (DG/UDA/TTA) and controlled comparisons [10]. As a result, it is often unclear, when comparing recent methods, whether reported gains come from handling weather-specific failures or from more general transfer effects.

In the past few years, evaluation has become more practical: several benchmarks now include adverse-condition data and controlled corruptions, making weather-related failures easier to measure in a more systematic way. This has fueled rapid progress in weather-focused LiDAR segmentation, but comparisons are still messy because papers often differ in (i) what target-domain information is allowed (DG/UDA/TTA), (ii) which part of the pipeline is changed (input, representation, or objective), and (iii) the evaluation protocol. In this survey, we organize the literature along three consistent axes—measurement mechanisms, learning setting, and intervention level—so that methods can be read and compared under aligned assumptions.

To make this fast-growing literature comparable and interpretable, we organize prior work using a two-axis taxonomy that is grounded in LiDAR measurement mechanisms under adverse weather. The first axis is the **learning setting**—domain generalization, unsupervised domain adaptation (UDA), and test-time adaptation (TTA)—which specifies **when** (if ever) target-domain scans are available and therefore what information an algorithm is allowed to use. Orthogonal to this, the second axis captures **where** a method intervenes in the perception pipeline: B1 input-space/preprocess techniques that modify the raw observations, B2 representation-space approaches that reshape intermediate features (e.g., cue decoupling and controlled collaboration, uncertainty/statistics calibration, or structure-preserving feature design), and B3 objective/prediction-space methods that impose robustness constraints or leverage target-side training signals. Importantly, this taxonomy is grounded in the sensing physics: because adverse weather affects geometry and radiometry differently, our categorization is not a mere enumeration, but a way to link design choices to the specific failure modes of weather-degraded LiDAR and to reason about why certain strategies transfer more reliably than others.

The contributions of this paper are as follows:

- We conduct a survey of adverse-weather LiDAR semantic segmentation for autonomous driving, and provide a protocol-aware framing (DG/UDA/TTA) that clarifies permissible target information and fair comparison rules.
- We provide a physics-grounded analysis of weather-induced domain shift, separating geometric degradation (e.g., clutter, truncation, wet-ground holes) from radiometric distortion (intensity drift), and summarize the key design implications for robust segmentation.
- We review representative weather-focused methods using a two-axis taxonomy—learning setting and pipeline intervention level (B1 input/preprocess, B2 representation, B3 objective/prediction)—and consolidate the datasets and evaluation metrics commonly used in this line of research.
- We compile and interpret reported cross-domain results on a common real-world adverse-weather target (SemanticSTF) under as consistent a backbone setting as possible, and summarize recurring failure modes, practical takeaways, and future research directions.

The remainder of this paper is organized as follows. Section 2 formalizes the task of point-wise LiDAR semantic segmentation and defines the learning settings (DG/UDA/TTA) used throughout the survey. Section 3 analyzes adverse-weather domain shift from a physics perspective, separating

geometric degradation from radiometric (intensity/return-strength) distortion and extracting design-relevant insights. Section 4 summarizes commonly used datasets and evaluation metrics for adverse-weather segmentation. Section 5 reviews representative methods using our two-axis taxonomy, organized by pipeline intervention level (B1 input/preprocess, B2 representation, and B3 objective/prediction) under different learning settings. Section 6 compiles reported cross-domain results, discusses empirical trends and failure modes, and highlights open directions for future research. Finally, Section 7 concludes the survey.

2. Problem Formulation and Learning Settings

This review focuses on the adverse-weather performance of LiDAR semantic segmentation for autonomous driving. In practice, the same segmentation model that performs well on clear-day data often degrades sharply in rain, fog, snow, or dust. This gap is not simply random noise: it is a systematic domain shift caused by weather-dependent point dropouts, backscatter, range-dependent attenuation, and intensity distortions. To make discussions and comparisons precise, we define the task and then distinguish learning settings by what information is available during training and deployment. This distinction directly determines what can be assumed, what is allowed, and what constitutes a fair evaluation.

2.1. Task Definition

We consider point-wise semantic segmentation on LiDAR point clouds. Let $P = \{(\mathbf{x}_i, \mathbf{f}_i)\}_{i=1}^{N}$ denote a LiDAR scan, where $\mathbf{x}_i \in \mathbb{R}^3$ is the 3D coordinate and $\mathbf{f}_i$ collects additional attributes (e.g., intensity, ring index, timestamp, or multi-return features when available). The goal is to predict a semantic label for each point:

$$\hat{y}_i = h_\theta(P)_i, \quad \hat{y}_i \in \{1, \dots, C\} \tag{1}$$

where h_θ is a segmentation model with parameters θ and C is the number of classes. Although we ultimately predict a semantic label for each point, different pipelines organize LiDAR measurements in different ways before feeding them into a network. In practice, this choice can matter in adverse weather: weather artifacts such as beam-wise dropouts, range-dependent sparsity, and intensity drift may be amplified (or partially smoothed out) by the intermediate representation and preprocessing. As a result, an adaptation technique that works well in one pipeline does not always transfer cleanly to another [65].

2.2. Learning Settings: What Is Available and When

To keep comparisons fair, we categorize methods by *when* target-domain information is accessible. We focus on three settings that are most relevant for autonomous driving deployment and widely used in the literature: domain generalization (DG), unsupervised domain adaptation (UDA), and test-time adaptation (TTA). Throughout, the supervision signal (semantic labels) is available only in the source domain unless stated otherwise.

2.2.1. Domain Generalization (DG, target-unseen)

In DG, we train the segmentation model only on labeled clear-weather data and **never use target-domain (adverse-weather) samples during training**. Let $\mathcal{D}^s$ and $\mathcal{D}^t$ denote the source (clear) and target (adverse) domains. Given labeled source data

$$\mathcal{S} = \{(P_i^s, Y_i^s)\}_{i=1}^{n^s} \tag{2}$$

we learn parameters θ by standard supervised training on the source:

$$\theta^* = \arg\min_\theta \frac{1}{n^s} \sum_{i=1}^{n^s} \ell\left(h_\theta(P_i^s),\, Y_i^s\right). \tag{3}$$

Evaluation is then performed directly on the adverse-weather domain, i.e., $P^t \sim \mathcal{D}^t$, without any target-domain adaptation. This setting is particularly relevant for autonomous driving, where collecting and annotating diverse adverse-weather data is often difficult, and deployment conditions may differ from what was seen during development.

2.2.2. Unsupervised Domain Adaptation (UDA, target-unlabeled)

In UDA, we still have labeled clear-weather data, but we also have access to **unlabeled scans collected in adverse weather**. Concretely, in addition to $\mathcal{S}$, we are given unlabeled target data

$$\mathcal{T} = \{P_j^t\}_{j=1}^{n^t}, \tag{4}$$

and the objective is to learn a model that performs well on the target domain by combining supervision from $\mathcal{S}$ with additional constraints derived from $\mathcal{T}$. A generic UDA objective can be written as

$$\theta^* = \arg\min_\theta \underbrace{\frac{1}{n^s} \sum_{i=1}^{n^s} \ell\left(h_\theta(P_i^s),\, Y_i^s\right)}_{\text{source supervision}} + \lambda \underbrace{\frac{1}{n^t} \sum_{j=1}^{n^t} \mathcal{R}\left(h_\theta(P_j^t)\right)}_{\text{target-driven regularization}}, \tag{5}$$

where $\mathcal{R}(\cdot)$ represents a target-side adaptation signal (e.g., consistency, entropy minimization, feature alignment, or pseudo-labeling), and λ balances the two terms.

In practice, UDA is implemented in two main ways. Offline UDA assumes the target data (or a large subset) is available during training, while online UDA adapts as target scans arrive over time, which is closer to deployment. Compared with DG, UDA often yields larger improvements, but it also makes evaluation easier to "game": performance can depend strongly on how much target data is used, how diverse it is, and whether hyper-parameters are implicitly tuned on the target domain.

2.2.3. Test-Time Adaptation (TTA)

TTA updates the model during deployment **using the incoming target-domain stream, typically without labels**. Compared with UDA, it is much more constrained: adaptation has to happen under tight latency, memory, and safety budgets, and it cannot rely on repeated access to a curated target dataset.

A convenient way to describe TTA is to start from a trained model θ_0 (from DG training, or after UDA) and apply a lightweight update rule when a target scan P_k^t arrives:

$$\theta_{k+1} = \mathcal{A}\big(\theta_k,\, P_k^t\big) \tag{6}$$

where $\mathcal{A}(\cdot)$ could mean updating a few parameters, refreshing normalization statistics, or optimizing a self-supervised objective such as prediction consistency, entropy regularization, or reconstruction.

TTA is especially relevant in adverse weather because conditions can shift within a single drive (e.g., clear → fog → spray), and because the "target domain" is rarely stationary—it is a moving mixture of weather severity, lighting, and scene content. This makes TTA particularly important for real deployments, where a segmentation model may need to adapt online to continuously changing observations rather than to a fixed target dataset prepared in advance. In this sense, TTA is more deployment-oriented than offline UDA: the challenge is not only to exploit target-domain information, but to do so under streaming inputs and strict latency, memory, and safety constraints. At the same time, TTA can be fragile. When the input is heavily corrupted, the self-supervised adaptation signal itself may become unreliable, causing error accumulation or model drift during operation.

3. Physics and Domain Shift Analysis under Adverse Weather

Adverse weather does not simply "add noise" to LiDAR point clouds. Rain, snow and fog change the full measurement process: photons are attenuated and scattered along the path, specular surfaces become wetter and more reflective, and the sensor itself may adapt its gain or discard weak returns. Classical analyses from automotive lidar physics already show that these mechanisms jointly affect range, intensity and detection probability in a highly distance- and wavelength-dependent way [44]. Recent simulation and robustness studies show that adverse weather introduces systematic, spatially correlated, and range-dependent perturbations changes in LiDAR point clouds, such as range-dependent sparsity, clutter, and intensity bias rather than simple noise that is independent across points and follows the same distribution everywhere [17, 16, 9, 27]. For LiDAR semantic segmentation in autonomous driving, it is therefore useful to distinguish two complementary views of the domain shift: (i) geometric-level effects, which reshape the spatial support and density of points, and (ii) intensity / measurement-level effects, which alter intensity distributions and the relative reliability of different beams or returns. The roadmap of this section is shown in Fig. 2

3.1. Geometric-level Effects

From a geometric point of view, adverse weather primarily changes which 3D locations produce valid returns and how densely they are sampled.

3.1.1. Weather-induced clutter and local geometry distortion

Precipitation (rain and snow) introduces additional scatterers along each beam, so part of the emitted energy is backscattered before reaching the true surface. When this backscattered return exceeds the sensor's detection threshold, it can produce spurious near-range points (often described as **ghost echoes** or **salt-and-pepper** clutter) that are easy to confuse with small objects in segmentation [18, 22, 52, 65, 27]. Monte-Carlo analyses further suggest that droplets close to the sensor are statistically more likely to trigger these artifacts due to the distance-dependent propagation loss [27]. In parallel, physics-based simulators explicitly sample weather particles and compute their interaction with LiDAR beams: the snowfall simulator generates dense "curtains" of near-range clutter together with missing returns on distant objects [16], while related simulators also model rain and mixed fog–rain conditions in a similar spirit [17, 27]. In real adverse-weather driving data (e.g., STF [3], SemanticSTF [57]), these effects appear as halos of points around headlights or in front of the ego vehicle, where the true geometry of cars and pedestrians is partially masked by weather clutter [65]. Overall, this clutter-and-occlusion pattern disrupts local geometric cues and scene continuity, which often leads clean-trained models to over-segment clutter regions and under-segment partially occluded road structure and obstacles.

3.1.2. Range-dependent attenuation and truncation

In fog (and, in severe cases, heavy rain), microscopic particles act as a scattering medium that dissipates laser energy along the propagation path. A standard physical description is exponential attenuation under the Beer-Lambert law, so the energy available for a valid surface return decreases rapidly with distance [17]. Once the reflected signal falls below the sensor's effective noise floor or detection threshold, the corresponding point is lost entirely, leading to range truncation and a pronounced sparsity increase in the far field. This mechanism explains why physics-based fog simulation produces point clouds where distant structures largely vanish and only a short corridor in front of the vehicle remains populated. From a robustness perspective, these effects align with corruption benchmarks reporting that **density decrease** and **field-of-view loss** are among the most harmful perturbations for LiDAR perception, especially under low-visibility conditions [44, 12, 19, 22, 17, 34, 29, 50]. For autonomous driving, this loss of far-range returns is not merely a metric drop: it removes the early, stable context that supports lane-level scene understanding and safe foresight. When

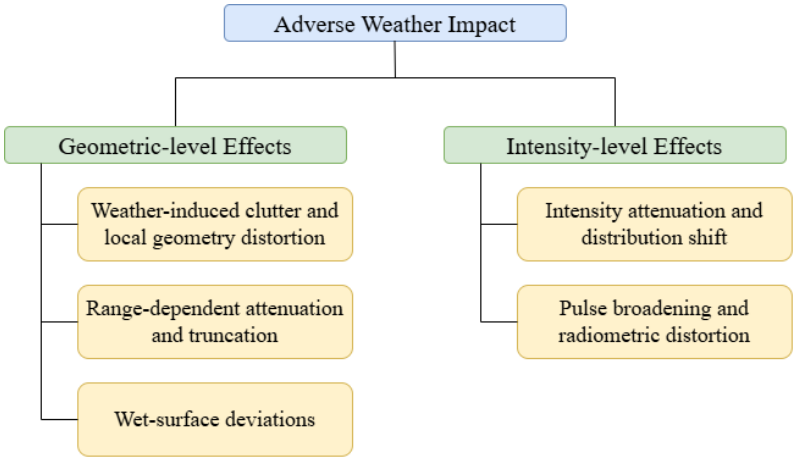


Figure 2: Weather effect on lidar point cloud.

distant road boundaries, curbs, buildings, and vegetation become sparse or disappear, segmentation models are forced to operate with a much shorter effective horizon, which can destabilize predictions and reduce reliability for downstream planning—especially at speed or near intersections where long-range scene structure normally provides critical cues.

3.1.3. Wet-surface deviations

Geometric shifts under adverse weather are not limited to airborne scatterers. After rainfall or snowfall, water films and slush can form on the road surface and change its reflection characteristics from largely diffuse to more specular. In this case, a substantial portion of the emitted energy is reflected away from the receiver rather than back toward it, which can lead to missing or weakened ground returns. Physics-based modeling and simulation studies show that this effect may create "black holes" on the ground plane—local patches where the road surface is under-sampled or disappears entirely in the point cloud due to specular reflection [17, 16, 60, 27]. For LiDAR semantic segmentation, these missing ground measurements are particularly problematic because the ground plane is a stable geometric reference used by many pipelines, either explicitly (ground removal and normalization) or implicitly (height-based features, range-image continuity, and contextual cues for road boundaries). When the ground becomes fragmented, models can confuse road with sidewalk or nearby low-lying objects, and boundary regions become noisy—exactly where autonomous driving needs reliable free-space estimation.

3.2. Intensity and Measurement-level Effects

While geometric shifts mainly change **where** points appear, intensity and measurement-level shifts change **what** information those points carry: the received signal strength (often recorded as intensity) and related sensor measurements provide appearance-like cues that are surprisingly close to semantic hints, much like in camera LiDAR fusion where LiDAR anchors geometry and the image contributes material and texture information.

3.2.1. Intensity attenuation and distribution shift

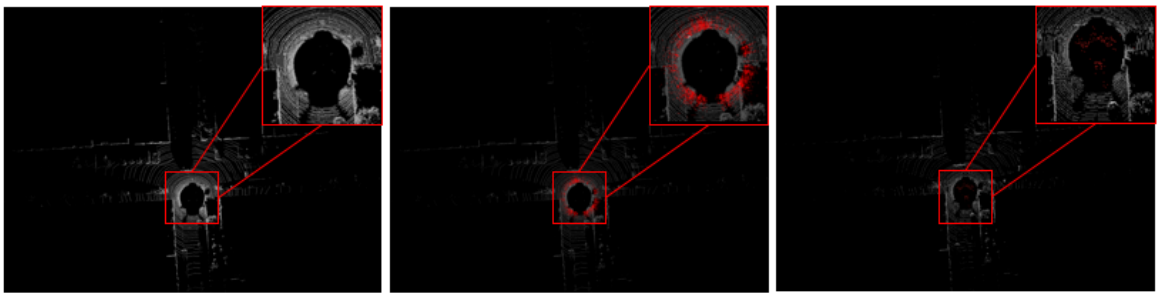
In LiDAR, the recorded intensity I is closely related to the received power P_R of a return. Under clear conditions, P_R is mainly governed by the target reflectivity ρ and the geometric range dependence (often simplified as an inverse-square law). In adverse weather, the signal must propagate through a scattering medium, so the atmospheric transmission decreases with distance due to extinction [17, 27]. A standard way to describe this attenuation is through the received power P_R as a time-wise convolution of the transmitted signal power P_T and the optical system's impulse response H [65]:

$$P_R(R) = C_A \int_0^{2R/c} P_T(t)\, H\left(R - \frac{ct}{2}\right) dt, \tag{7}$$

where R is the object distance, c is the speed of light, and C_A is a system constant independent of time and range. In adverse weather, scattering and extinction effectively reduce the energy that reaches the target and returns to the receiver, which lowers P_R and, in turn, compresses the observed intensity distribution—especially at longer ranges. The practical implication is that, even for the same material at the same distance, the sensor can observe systematically weaker returns in fog or heavy rain than in clear weather [18, 12, 19, 22, 25, 15].

3.2.2. Pulse broadening and radiometric distortion

Beyond simple attenuation, adverse weather can distort the **shape** of the return pulse, which changes not only intensity but also the reliability of range and material cues. In fog, the received signal can be viewed as a superposition of a hard-target echo from the actual surface and a distributed soft-target echo caused by volumetric backscattering in the fog medium [18, 22, 17]. This superposition broadens the pulse and may shift the detected peak, leading to increased ranging uncertainty even when an object is nominally within sensing distance. In addition, surface wetness introduces thin-film effects on the road surface: the effective reflectivity can become a function of water-layer thickness and



(a) Clear (b) Fog (c) Rain

Figure 3: Geometry-level weather degradation in fog and rain (clutter and range loss). Bird's-eye-view visualization of a clear LiDAR scan and its fog- and rain-degraded counterparts generated by widely used physics-based simulation studies [17, 27].

incidence angle, producing intensity fluctuations that are difficult to predict from geometry alone [16]. Together, these distortions make intensity-based appearance cues unreliable across weather unless reliability is modeled.

3.3. Domain-Shift Insights and Design Implications

The above analysis highlights a domain shift that is both **structured** and **asymmetric**: adverse weather does not typically change the driving scene itself, but it changes how the scene is observed. On the geometry side, the dominant layout (road topology, major surfaces, and large structures) is often still present, yet the evidence becomes **locally fragmented**—near range is contaminated by clutter and partial occlusion, far range is truncated by signal extinction, and the ground plane can develop discontinuous holes due to wet-surface reflection (Sec. 3.1.1–3.1.3). This local degradation with global similarity viewpoint is important for autonomous driving: it shortens the effective sensing horizon and reduces the stability of lane-level context, which in turn increases uncertainty exactly when downstream planning needs reliable free-space and boundary cues. This behavior is also evident in Fig. 3: compared with the clear scan, fog and rain introduce a compact band of near-range clutter (red points) while progressively erasing far-field structures, so the overall scene layout remains recognizable but the available geometric evidence becomes uneven across space.

To quantify the asymmetric weather-induced shift at the dataset level, we plot frame-balanced empirical cumulative distribution functions (ECDFs) of point distance and intensity across clear and adverse-weather domains. Fig. 4 shows that the distance ECDFs for SemanticKITTI (clear), SemanticSTF (real adverse weather), and weather-corrupted SemanticKITTI-C largely overlap, suggesting that the **global** range distribution is only mildly perturbed when aggregated over many frames (even though individual scenes can exhibit severe local clutter and far-range truncation). In contrast, the intensity ECDFs diverge substantially across domains: fog/snow corruptions shift mass toward low intensity values, whereas SemanticSTF exhibits a broader and less calibrated distribution. This dataset-level evidence supports a key insight: geometry provides a comparatively stable anchor under weather, while intensity is a more condition-dependent and thus riskier cue.

This instability of intensity is expected from the underlying measurement physics. Under fog, rain, or snow, extinction and backscattering attenuate returns and compress intensity statistics; wet surfaces and pulse distortion further reduce intensity reliability (Sec. 3.2.1–3.2.2). Consequently, the same scene may present markedly different appearance-like signals across weather, so models that heavily rely on intensity patterns learned in clear conditions can become systematically miscalibrated. Fig. 5 provides an example at the frame level: in fog, the range-view intensity map collapses toward low values with sparse high-intensity outliers, indicating reduced radiometric contrast and unreliable spikes; rain is less extreme but still shifts and smooths intensity patterns, weakening material separability. These observations also explain why removing intensity can sometimes improve generalization across adverse conditions [61]: intensity is beneficial only when its calibration and local consistency are trustworthy, motivating reliability-aware designs that gate or calibrate intensity features rather than treating them as universally stable inputs.

These observations suggest practical design implications for domain adaptation methods under adverse weather. First, instead of pushing for global feature alignment, it is often more effective to encode **reliability-aware** representations: models should distinguish "missing/uncertain evidence" (due to truncation or wet-ground holes) from true semantic changes, so that clutter is less likely to be mistaken as objects and empty regions are not over-confidently labeled. Second, geometry can be treated as an **anchor** for cross-domain consistency, while intensity should be treated as a **selectively trusted** cue, for example, by calibrating, gating, or conditioning intensity features on geometric context or estimated measurement quality.

4. Datasets and metrics

In this section, we summarize the datasets and evaluation metrics that are commonly used to study the adaptation ability of point-cloud semantic segmentation models in autonomous driving under adverse weather. We first review

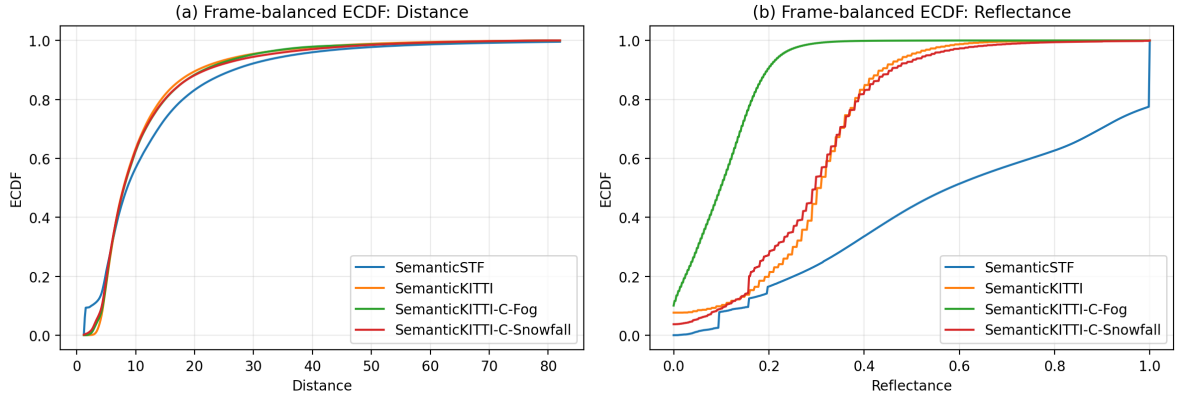


Figure 4: Frame-balanced ECDF comparison of geometry and intensity across domains. (a) ECDF of point distance to the LiDAR sensor. (b) ECDF of LiDAR intensity (normalized). The y-axis reports the empirical cumulative probability, i.e., the fraction of points with value $\leq x$. We adopt a frame-balanced ECDF so that each scan contributes equally, avoiding bias from varying point counts across frames. The graph (b) indicates that fog significantly affects intensity: nearly 80% of the data distribution has a normalized intensity of ≤ 0.2.

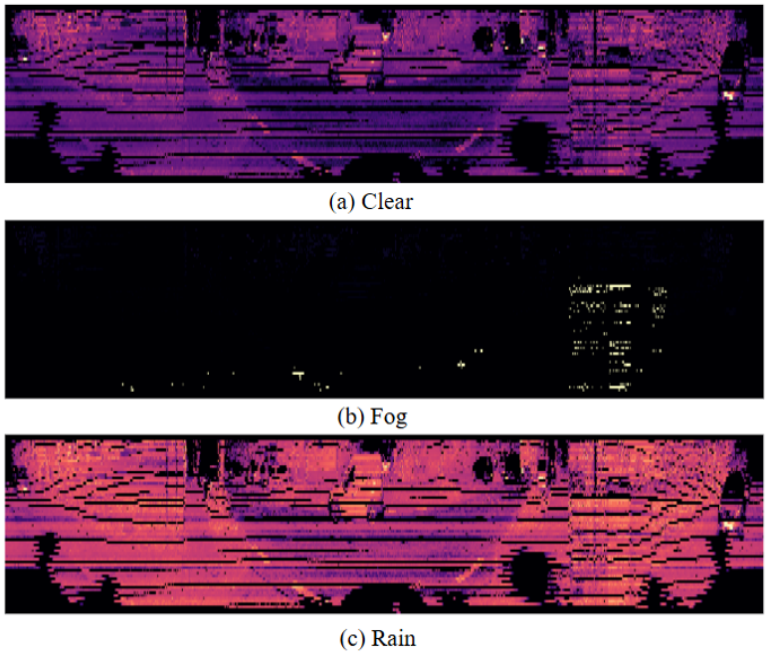


Figure 5: Measurement-level intensity shift in fog and rain. Range-view (spherical projection) visualizations of the LiDAR intensity channel for a clear scan and its simulated fog-/rain-degraded counterparts [17, 27]. Colors represent normalized return intensity in $[0, 1]$ with a shared scale (magma); dark pixels indicate low intensity or empty projections (no return), and bright pixels indicate stronger returns.

outdoor LiDAR-based datasets that provide point-wise semantic (or panoptic) annotations, and categorize them by data source (real-world collection vs. simulation/synthesis) and weather coverage (clear vs. adverse conditions such as fog, rain, and snow), since these factors directly determine the domain shift and the difficulty of adaptation. Datasets that do not support point-level semantic segmentation, as well as indoor datasets that are not representative of driving environments, are outside the scope of this survey. We then introduce the standard metrics and evaluation protocols (e.g., mIoU and robustness-oriented reporting across weather domains) used to quantify segmentation accuracy and generalization.

4.1. Datasets

SemanticKITTI [2] is a large-scale, real-world LiDAR semantic segmentation dataset built on the KITTI Vision Benchmark odometry sequences. It covers diverse outdoor driving scenes around Karlsruhe, Germany. Following the original KITTI odometry split, it uses sequences 00–10 for training and 11–21 for testing, providing dense point-wise semantic annotations for each full 3D scan over the

LiDAR's full surround field-of-view, making it a widely adopted benchmark in autonomous driving perception.

Beyond its scale, SemanticKITTI is designed for sequential LiDAR understanding and supports multiple benchmarks, including single-scan semantic segmentation, multi-scan (temporal) semantic segmentation, and semantic scene completion. It defines 28 annotated semantic classes (with moving/non-moving variants for several dynamic categories and an outlier label), while many works adopt a reduced label set by ignoring rare labels and/or merging moving classes into their static counterparts. In the context of adverse-weather adaptation for autonomous-driving 3D semantic segmentation, SemanticKITTI is most frequently used as the source-domain dataset for training under (relatively) clear-weather conditions, and models are then evaluated on adverse-weather target datasets to quantify the domain shift.

SynLiDAR [56] is a large-scale synthetic sequential LiDAR dataset for outdoor point-cloud semantic segmentation in autonomous driving. It is generated in Unreal Engine 4 (UE4) with carefully modeled driving environments (e.g., urban streets and other outdoor traffic scenes), which enables automatic collection of LiDAR point clouds together with dense point-wise semantic labels. To further reduce the synthetic-to-real gap on the sensing side, SynLiDAR also provides an intensity modeling strategy by predicting more realistic intensity values for synthetic points using a model trained on real LiDAR data.

In terms of scale and annotations, SynLiDAR contains 13 sequences with 198,396 scans and point-wise labels for 32 semantic classes. In adverse-weather adaptation or domain generalization studies for autonomous-driving 3D semantic segmentation, SynLiDAR is often used as a source-domain dataset for training or pretraining: models learn from fully labeled synthetic data (typically treated as clear-like source data) and are then transferred to real-world datasets—including adverse-weather benchmarks—to evaluate robustness and domain shift.

SemanticSTF [57] is a real-world outdoor LiDAR semantic segmentation dataset built upon the STF Detection Benchmark [3]. It is collected across multiple European countries (e.g., Germany, Sweden, Denmark, and Finland) using a Velodyne HDL64 S3D LiDAR, and is curated to cover diverse adverse-weather driving scenes. In total, SemanticSTF contains 2,076 LiDAR scans under challenging conditions, including snow, dense fog, light fog, and rain, and is split into 1,326/250/500 scans for train/validation/test with similar weather proportions across splits.

SemanticKITTI-C [59] is a robustness benchmark for LiDAR semantic segmentation that extends SemanticKITTI by systematically injecting controlled corruptions into the original point clouds. It is designed to emulate a range of real-world sensing degradations and domain shifts, and organizes the perturbations into three main corruption domains: adverse weather, measurement noise, and cross-device discrepancy. Concretely, SemanticKITTI-C defines several corruption types with multiple severity levels, including fog and snowfall simulations, global outliers, local geometric distortions, and reduced-beam configurations (e.g., 64-beam and 16-beam variants), which together enable fine-grained stress testing of LiDAR segmentation models under progressively more challenging conditions.

SemanticSTF provides high-quality point-wise annotations for adverse-weather point clouds and follows a SemanticKITTI-compatible class taxonomy for evaluation. Besides the standard semantic categories, it additionally marks weather-induced indiscernible regions (e.g., heavy snow coverage) as *invalid* and ignores unlabeled or ambiguous points during training and evaluation. Together with SemanticKITTI-C, which injects synthetic fog, snow, noise, and cross-device corruptions on top of clean SemanticKITTI scans, these two benchmarks provide complementary ways to assess adverse-weather robustness: SemanticSTF is widely used as a real-world **target-domain** dataset to evaluate how well models trained on clear-weather sources transfer to fog/rain/snow conditions, while Semantic KITTI-C serves as a controlled corruption benchmark for quantifying performance degradation under systematically varied weather-like distortions.

Beyond the representative datasets discussed above (SemanticKITTI, SynLiDAR, SemanticSTF, and SemanticKITTI-C), a variety of other outdoor LiDAR benchmarks can also be exploited to study adverse-weather adaptation for autonomous-driving semantic segmentation. As summarized in Table 1, these include richly annotated clear-weather datasets such as KITTI-360 [33], nuScenes / Panoptic-nuScenes [13], the Waymo Open Dataset [51], PandaSet [58], A2D2 [14], SemanticPOSS [37], SemanticUSL [24], and MLDAS [5], as well as real-world adverse-weather datasets like WADS [31] and SemanticSpray [40]. Together with additional synthetic or simulation-based resources such as PreSIL [23], this collection of datasets spans real versus synthetic sources, clear versus adverse conditions, and single- versus multi-LiDAR settings, providing flexible source–target combinations for evaluating and comparing domain adaptation and generalization methods under weather-induced distribution shifts.

4.2. Metrics

For LiDAR semantic segmentation, the most commonly used evaluation metric is class-wise Intersection-over-Union (IoU), and the overall performance is typically summarized by mean IoU (mIoU), i.e., the average IoU across the evaluated semantic classes. Intuitively, IoU measures how well the predicted points of a class overlap with the ground-truth points of that class: a higher IoU means that the model captures that semantic region more accurately, with fewer missed detections and fewer incorrect predictions. Correspondingly, a higher mIoU indicates that the segmentation model achieves better overall agreement with the ground truth across classes, while a lower mIoU reflects more severe confusion, omission, or boundary errors in prediction.

Because mIoU balances performance across both frequent and rare categories, it has become the standard metric

Table 1
Outdoor LiDAR datasets for point-cloud semantic segmentation under clear/adverse weather.

Dataset	Source	Weather	Classes	Description
SemanticKITTI [2]	Real world	Clear	28	Standard clear-weather baseline
KITTI-360 [33]	Real world	Clear	37	Richly annotated suburban driving
Panoptic-NuScenes [13]	Real world	Mixed (rain)	32	Dense urban panoptic
WOD [51]	Real world	Mixed (rain/fog)	23	Large-scale diverse urban/suburban
PandaSet [58]	Real world	Clear	37	Complex urban environments
A2D2 [14]	Real world	Mixed (rain)	38	Commercial license / Urban, country, highway
SemanticPOSS [37]	Real world	Clear	14	Dynamic campus scenes
SemanticUSL [24]	Real world	Clear	28	Campus / Off-road
MLDAS [5]	Real world	Clear	14	Multi-LiDAR (32/64/128 beam)
SemanticSTF [57]	Real world	Adverse (rain/fog/snow)	21	Adverse weather benchmark (fog/rain/snow)
WADS [31]	Real world	Adverse (snow)	22	Severe winter / Dense point-wise labels
SemanticSpray [40]	Real world	Adverse (rain)	3	Vehicle spray on wet surface
Panoptic-CUDAL [54]	Real world	Adverse (rain)	19	Rural Australia / Rainy / Panoptic labels
REHEARSE-3D [43]	Real world	Adverse (rain)	8	Real-world emulated rain / De-raining
SynLiDAR [56]	Synthetic	Clear	32	Large-scale synthetic / Sim-to-Real
PreSIL [23]	Synthetic	Clear	2	Sim-to-Real
SemanticKITTI-C [59]	Synthetic	Adverse (fog/snow)	19	Synthetic corruptions
Weather-KITTI [66]	Synthetic	Adverse (rain/fog/snow)	21	Synthetic adverse weather
Weather-NuScenes [66]	Synthetic	Adverse (rain/fog/snow)	19	Synthetic adverse weather based on NuScenes
SnowyKITTI [49]	Synthetic	Adverse (snow)	2	Denoising benchmark
Occluded-NuScenes [30]	Synthetic	Adverse (rain)	32	Synthetic Multi-sensor Occlusions

for benchmarking LiDAR semantic segmentation, including studies on robustness under adverse weather. In this survey, however, the metric itself is only one part of fair comparison. The interpretation of reported mIoU also depends on the underlying evaluation protocol, such as the label set, whether moving classes are merged, and how invalid or ambiguous points are handled. For this reason, when we later compare cross-domain results on SemanticSTF in Table 5, we place particular emphasis not only on the reported mIoU values, but also on keeping the evaluation setting as consistent as possible across papers.

5. Adverse-Weather Adaptation for 3D LiDAR Semantic Segmentation

Building on the problem formulation, physical analysis, and datasets summarized in Sections 2–3.3, this section reviews methods that improve the adverse-weather robustness of LiDAR semantic segmentation in autonomous driving. Our goal is not only to list individual techniques, but to understand **where** in the perception pipeline they intervene and **which** learning protocol they assume. To this end, we organize the literature using a two-level taxonomy. At the first level, each method is tagged by its learning setting—domain generalization (DG), unsupervised domain adaptation (UDA), or test-time adaptation (TTA)—as defined in Section 2. At the second level, we classify methods by their primary intervention position in the LiDAR processing pipeline: input-space / preprocess (B1), representation-space (B2), and objective / prediction-space (B3). The remainder of this section follows this two-level view when grouping and comparing existing work. Although TTA is highly relevant for real-world deployment under continuously changing weather, weather-specific TTA methods for LiDAR semantic segmentation are still relatively limited compared with DG and UDA.

Our scope is deliberately restricted to methods that *explicitly* address adverse-weather effects. In particular, we include approaches whose design or evaluation directly involves weather-induced degradations such as fog, rain, snow, spray, or their controlled corruption counterparts (e.g., SemanticSTF[57], SemanticKITTI-C[59], Weather-KITTI[66]). Conversely, we do not aim to cover the much broader body of 3D domain adaptation or domain generalization methods that are only validated under clear-weather cross-dataset shifts, such as generic range-view mixing or feature-alignment techniques (e.g., PolarMix[55], CoSMix[46], and other works[28, 48, 47, 45]) trained from SemanticKITTI to nuScenes. While these works are important for handling changes in traffic scenes, sensors, or annotation policies, they typically do not model the characteristic geometry and measurement level shifts caused by adverse weather, namely near range clutter, range truncation, wet ground "holes", and intensity compression or pulse distortions summarized in Section 3. Including them would blur the focus of this review and make it harder to isolate what truly improves LiDAR semantic segmentation under physically grounded weather corruptions. We also observe an interesting temporal pattern: weather-focused 3D LiDAR semantic segmentation adaptation papers only start to appear in large numbers from 2024 onward, which is unsurprising given that the community largely lacked large-scale, publicly available adverse-weather segmentation benchmarks before 2023, and datasets such as SemanticSTF [57] have only recently made systematic, weather-specific adaptation studies feasible.

At the first level of our taxonomy, we tag each method by its learning setting—domain generalization (DG), unsupervised domain adaptation (UDA), or test-time adaptation (TTA)—following the definitions in Section 2. DG methods are trained only on labeled clear-weather data and are evaluated directly on adverse-weather domains without seeing target scans during training. UDA methods still rely on labeled clear-weather sources, but additionally exploit unlabeled target-weather scans, either in an offline fashion (using a fixed pool of target data) or in an online fashion (adapting as target streams arrive). TTA methods start from a pre-trained model (from DG or UDA) and update a subset of parameters on-the-fly during deployment, driven by self-supervised signals on the incoming target stream and subject to strict latency and safety constraints. We treat this axis as a fundamental descriptor because it determines what information is available to the algorithm and what constitutes a fair comparison: a DG method that never accesses target data and a UDA method tuned with abundant target examples solve related but distinct problems, even if they operate at the same intervention level in the pipeline.

Orthogonal to the learning setting, we categorize methods by **where** they intervene in the LiDAR perception pipeline. We adopt three intervention levels. B1 (input-space / preprocess) denotes methods that act directly on the raw observations, for example by simulating or removing weather artifacts through point-level corruptions, denoising, reflectance calibration, or beam-wise inpainting before the data enters the backbone. B2 (representation-space) covers approaches that redesign or augment the feature representation, such as separating geometry and reflectance branches, learning weather-robust latent spaces, or introducing reliability-aware fusion modules that reshape intermediate features without necessarily changing the input. B3 (objective / prediction-space) refers to methods whose main novelty lies in the training signal, including consistency and entropy-based regularizers across weather conditions, pseudo-label and self-training schemes on adverse-weather scans, or uncertainty-aware and reconstruction losses coupled with the segmentation objective. For each paper, we assign a single **primary** B-tag based on where its core novelty and dominant gains are introduced. We label a method as **B1** if it primarily modifies the input/observations before feature extraction, **B2** if it mainly redesigns intermediate representations within the network, and **B3** if it is driven chiefly by objective- or prediction-level constraints (e.g., consistency, entropy/risk reweighting, or pseudo-supervision). When a paper combines multiple components, we keep one primary tag for comparability and note secondary components in the text where relevant. The roadmap of this section is shown in Fig. 6.

5.1. Weather-Aware Input Preprocessing (B1)

B1 methods improve adverse-weather robustness by acting on the **observations** before feature extraction. They modify the raw point cloud (or its input representation such as range-view) rather than changing the backbone or the loss. This design follows the physics of weather-induced domain shift (Sec. 3.3): precipitation creates near-range clutter, fog/heavy rain causes range truncation and beam-wise sparsity, wet surfaces lead to local ground "holes", and radiometric effects distort intensity statistics. We group B1 techniques into two families. The first applies **weather-aware input augmentation** directly to the original scans, using corruptions that correspond to these weather effects so the model is trained on target-like observations. The second performs **weather mixing**, which composes clean scans with weather artifacts in a controlled manner to form partially corrupted but label-consistent inputs. These input-level methods are usually model-agnostic and easy to use under DG/UDA/TTA, but their gains are bounded when weather removes information or alters radiometry in a strongly condition-dependent way. Table 2 summarizes representative B1 papers by learning setting and the specific input operations they apply.

A related B1 line performs LiDAR denoising or weather removal (e.g., de-raining [42] / de-snowing [63, 1] / de-fogging [64, 36]) before segmentation. We do not emphasize this direction for two reasons: (i) in heavy fog or long-range attenuation the main issue is *missing returns* rather than additive noise, so restoration can easily hallucinate structure; (ii) many denoising works are evaluated with reconstruction-oriented metrics or sensor-specific settings, making fair comparison under DG/UDA/TTA segmentation protocols difficult. We therefore only discuss denoising when it is directly paired with segmentation and validated on adverse-weather segmentation benchmarks.

5.1.1. Weather-aware input augmentation

Weather synthesis and domain randomization form a core B1 strategy: instead of changing the backbone or training objective, these methods *expand the input distribution* so that adverse-weather artifacts are already present during training. This direction matches the domain-shift analysis in Sec. 3.3: the semantics of driving scenes remain largely unchanged, while the LiDAR observation process is altered by structured effects such as backscatter clutter, range-dependent truncation, wet-surface dropouts, and intensity drift.

Dabbiru et al. [8] generates synthetic LiDAR point clouds with semantic labels under adverse weather using a physics-based simulator (MAVS). By explicitly varying weather severity (rain rate and dust opacity), they create "weathered" scans and train a standard range-view segmentation pipeline on these data to study and improve robustness under rain and dust conditions.

Kang et al. [26] propose LAWA, a physics-inspired adverse-weather augmentation pipeline that transforms clear-weather LiDAR scans into rain/snow-like observations for training robust point-cloud semantic segmentation models. Concretely, LAWA combines a scattering-point generation module with a wet-ground module to emulate precipitation-induced clutter, intensity attenuation, and distance-dependent

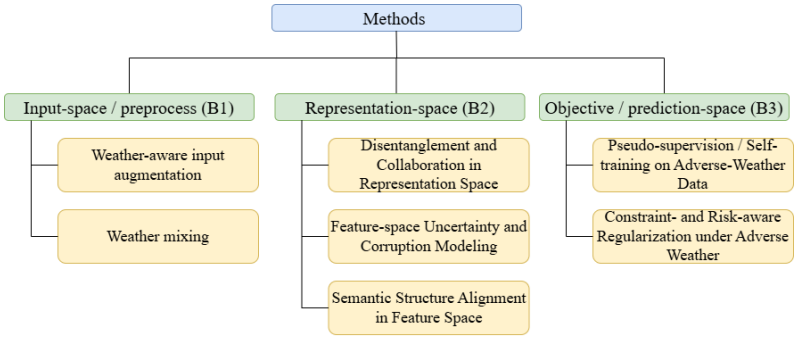


Figure 6: The proposed taxonomy of existing methods.

Table 2
B1 (Input-space / Preprocess) methods for adverse-weather LiDAR semantic segmentation.

Setting	Reference	Year	Weather focus	Input operation
DG	PointDR [57]	2023	fog/rain/snow	Geometry style randomization (rotation/scaling + dropout/noise/jitter)
DG	Park et al. [38]	2024	fog/rain/snow	Selective Jittering (depth/angle/range) + Learnable Point Drop (DQN-based)
DG/UDA	UniMix [65]	2024	fog/rain/snow	Bridge domain via physics-based weather simulation + Universal Mixing (spatial/intensity/semantic)
DG	Liu et al. [35]	2024	fog	Physics-based fog simulation (Foggy SemanticKITTI) + clustering-based outlier removal
DG	LAWA [26]	2025	rain/snow	Scattering-point synthesis + wet-ground modeling (intensity attenuation + range noise)
DG	Dabbiru et al. [8]	2025	rain/dust	MAVS physics-based simulation for adverse-weather LiDAR (auto labels)

noise (including wet-surface artifacts), and uses the resulting augmented scans to construct a weathered version of SemanticKITTI (with an extra scattering class) for training and evaluation under adverse conditions [26].

Xiao et al. [57] propose PointDR as an all-weather DG baseline on SemanticSTF by applying **geometry style randomization** to clear-weather point clouds, forming weak/strong augmented views via spatial transforms and stronger perturbations such as dropout, noise, and jitter. They then adopt an embedding aggregation contrastive framework, where point features are pulled toward class-wise prototypes maintained in a memory bank, encouraging weather-robust representations under geometry perturbations.

Park et al. [38] introduce an interesting data-centric augmentation strategy that uses reinforcement learning to learn where to drop points during training. Instead of relying on hand-crafted dropout rules, they train a DQN-based policy (Learnable Point Drop) to generate challenging point-removal patterns, and optimize the segmenter on these harder inputs so it becomes less sensitive to beam-wise sparsity and weather-induced occlusions [38]. The key takeaway for adverse-weather robustness is that input augmentation can be made adaptive: the worst-case drop patterns are discovered automatically and used as training signals, rather than being fixed by manual design.

Weather-aware input augmentation tackles adverse-weather shift by moving robustness "upstream": it makes the training data itself reflect the measurement degradations the model will face at test time. Physics- and simulator-driven pipelines provide targeted weather effects with controllable severity, while domain randomization trades fidelity for broader coverage of geometric corruption patterns. An emerging insight is that augmentation need not be hand-designed: RL-based point dropping learns hard sparsity patterns from the model's own failures and turns them into training signals. Together, these methods suggest a practical recipe for DG under weather: prioritize realistic corruption coverage in the input space, and use adaptive augmentation when the dominant failure mode is missing or unreliable returns.

5.1.2. Weather mixing

Weather mixing improves adverse-weather robustness by **composing** weather artifacts with otherwise clean scans, rather than generating fully synthetic scenes. Mixing-style augmentation is widely used in 3D adaptation and generalization (e.g., PolarMix [55], CoSMix [46]), but weather-oriented mixing differs in that it explicitly targets the measurement degradations induced by rain/fog/snow rather than generic cross-dataset shifts. Compared with global corruptions, weather mixing injects degradations in a controlled and spatially structured way that better reflects real weather

signatures, such as near-range clutter bands, sector-wise sparsity, and range-dependent intensity shifts (Sec. 3). An additional benefit is semantic preservation: the underlying scene layout and labels remain unchanged, while only the observation is partially corrupted. As a result, weather mixing can produce diverse, label-consistent training samples that expose the model to realistic local degradations without requiring large-scale labeled adverse-weather data.

Zhao et al. [65] propose UniMix, a clear-to-adverse framework that treats weather shift as an input-space composition problem. They first build a simulated adverse "bridge" domain from clear scans via physically valid weather simulation, then apply a universal mixing operator to blend samples with masks defined over **spatial**, **intensity**, and **semantic** partitions, producing intermediate point clouds whose layouts remain consistent while weather signatures vary locally [65]. UniMix further adopts a two-stage teacher–student pipeline (with an EMA teacher) and can operate in both UDA and DG by reusing the stage-1 model when target data is unavailable [65]. A useful takeaway is that, for adverse weather, mixing becomes more than generic data interpolation: by explicitly controlling **where** corruption is applied and **which** attributes are perturbed (geometry vs. intensity), it provides a practical way to generate realistic partial degradations while keeping semantics stable, which is exactly the failure mode described in Sec. 3.

Overall, weather mixing offers a convenient middle ground between fully synthetic generation and simple global corruptions. By composing weather artifacts locally while keeping the underlying semantics fixed, it produces label-consistent training samples that match the structured nature of adverse-weather degradation.

5.1.3. Summary of B1

B1 methods act on the **input** before feature extraction, so they are largely model-agnostic and easy to plug into DG/UDA/TTA. Their key benefit is practicality: weather-aware corruptions and simulators can mimic major measurement effects with controllable severity, while weather mixing creates label-consistent, partially corrupted samples without fully synthetic scenes or labeled adverse-weather data. Their limitations are that performance depends on the realism and coverage of the injected artifacts. Mismatch to real sensors, severities, and scenes can limit transfer, and input-level changes alone do not ensure reliability-aware features under heavy corruption.

5.2. Weather-Robust Representation Learning (B2)

B2 methods improve adverse-weather robustness by intervening in the **representation space**, i.e., by redesigning how intermediate features are constructed, calibrated, and organized inside the network. Compared with B1 input preprocessing, B2 does not directly manipulate raw observations; instead, it aims to make the learned features more resistant to weather-induced corruption and more informative for semantics under distribution shift. In this survey, we further divide B2 into three sub-categories according to the primary representation-level mechanism. **Disentanglement** factorizes LiDAR cues (e.g., geometry and reflectance/radiometry, or multi-view encodings) into separate feature streams and then performs controlled collaboration to reduce cross-cue interference. **Feature-space uncertainty modeling** explicitly captures weather-induced perturbations as uncertainty or corruption in feature space, enabling reliability-aware calibration such as down-weighting unreliable activations or stabilizing feature statistics. **Semantic structure alignment** reshapes the geometry of the feature space so that semantic structure is preserved across weather domains, for example by binding corrupted features to clean prototypes, suppressing class overlap, or enforcing discriminative class-wise organization. In the remainder of this section, we review representative methods under these three directions and discuss their respective trade-offs. Table 3 summarizes representative B2 papers by learning setting and the specific representation operations they apply.

5.2.1. Disentanglement and Collaboration in Representation Space

Methods in this group improve adverse-weather robustness by **decoupling** LiDAR representations into complementary feature streams and then **fusing them with explicit, controlled interactions**. The underlying idea is simple: weather does not corrupt all cues in the same way or at the same scale (as shown in Fig. 4), and early entanglement can let corrupted signals dominate and spread through the backbone. A common design is therefore to separate heterogeneous cues (e.g., geometry vs. intensity, or voxel-based vs. range-view encodings), let each branch learn cue-specific invariances, and only introduce cross-branch exchange through dedicated fusion blocks (e.g., gated fusion, cross-attention, or staged local–global collaboration). This separation–then–collaboration pattern helps retain complementary information when it is informative, while limiting cross-cue interference when one stream becomes unreliable, leading to more stable features without changing the raw input.

GRC [61] is one of the first weather-focused segmentation works that treats geometry–reflectance decoupling as the starting point. The authors first analyze cross-weather statistics and show an asymmetric shift: the overall distance distribution changes only mildly, while the reflectance/intensity distribution drifts much more. This suggests that naively mixing intensity with geometry can make the representation brittle under adverse weather. Motivated by this, they propose a Geometry–Reflectance Collaboration framework with two separate branches (geometry and reflectance), followed by a robust multi-level fusion module that combines local and global interactions to retain complementary cues while suppressing unreliable ones. On the standard SemanticKITTI→SemanticSTF benchmark, this design yields a clear improvement. An ablation makes the point even more concrete: for a sparse-conv baseline (MinkNet [7]), simply *dropping the intensity input* already produces a

Table 3
B2 (Representation-space) methods for adverse-weather LiDAR semantic segmentation.

Setting	Reference	Year	Weather focus	Representation operation
DG	DGUIL [20]	2024	fog/rain/snow	Feature-space uncertainty augmentation (sampling uncertain feature distributions) + momentum introspective learning for generalized embeddings
DG	Liao et al. [32]	2024	fog/rain/snow	Point Attention Block inserted into a PointDR-style pipeline to strengthen discriminative feature extraction under corruption
DG	He et al. [21]	2025	fog/rain/snow	Category-level Geometry Embedding + Geometric Consistent Learning
DG	GRC [61]	2025	fog/rain/snow	Geometry–intensity dual-branch encoding + robust multi-level collaboration (information constraint + local/global fusion)
DG	GRV [62]	2025	fog/rain/snow	Range-view stem decoupling (geometry vs. intensity) + geometric abnormality suppression + intensity distortion calibration
DG	NTN [39]	2025	fog/rain/snow	Semantic structure regularization for things: superclass prototype binding + beam-wise feature distillation (clean vs. corrupted alignment)
UDA	Chen et al. [4]	2025	snow/fog/rain	Overlap-aware alignment in feature space: invertible attention alignment with overlap suppression + quality-guided contrastive memory bank
TTA	NormFusion [67]	2025	fog/snow/wet ground	Stage-wise BN-statistics fusion (test-time stats early, training stats late) for real-time feature calibration

surprisingly large gain (e.g., 24.4→37.3 mIoU), turning "geometry-only" into a strong baseline that can match or outperform many heavier schemes. The takeaway is that intensity is often informative, but it becomes fragile under weather, so it is better handled in a decoupled and controlled way rather than being entangled with geometry from the outset.

Building on this decoupling intuition, GRV [62] revisits the same idea for **range-view** segmentation, where weather artifacts can be amplified after spherical projection. Instead of redesigning the whole network, they make a small, plug-and-play change at the front end: the stem is refactored into a geometry branch and a reflectance branch, and each branch is cleaned up before fusion. Concretely, a Geometric Abnormality Suppression (GAS) module down-weights weather-corrupted geometric responses, while a Reflectance Distortion Calibration (RDC) module uses a memory-guided AdaIN-style normalization to calibrate reflectance features. The processed features are then fused and fed into the original backbone and head, keeping the rest of the pipeline intact and the overhead low.

Taken together, these two lines of work make a consistent case for separate first, fuse later under adverse weather. Explicitly decoupling heterogeneous cues prevents early feature contamination, while structured fusion provides a controlled way to re-introduce complementary information when it is reliable. In practice, the strongest designs are not those that always trust reflectance, but those that treat it as a conditional cue and regulate its influence through dedicated collaboration blocks rather than naive concatenation or summation.

5.2.2. Feature-space Uncertainty and Corruption Modeling

Methods in this group model adverse-weather effects **in feature space** rather than on raw points. Compared with B1-style input simulation (which tries to reproduce specific sensing artifacts), feature-space modeling targets what the network actually uses after early encoding. In practice, it serves two closely related purposes. First, it **exposes the model to feature perturbations** that mimic weather-induced unreliability, so the backbone learns representations that are less sensitive to corrupted evidence (e.g., injecting feature noise sampled from a random/learned distribution, or perturbing intermediate features in a controlled way [62, 20]). Second, it **reduces or adapts to feature distribution shifts** between clear and adverse weather, for example by calibrating feature statistics with normalization-based mechanisms [20, 67], or by using a variational re-parameterization (VAE/VIB-style sampling [61]) to pull features into a shared latent space with a distribution regularizer. Both aim to keep intermediate features stable enough that later layers do not amplify local corruption into global prediction drift.

DGUIL [20] frames domain shift as **uncertainty in feature space**. During training, it deliberately perturbs point features to sample multiple plausible feature distributions, so the backbone cannot rely on a single clean representation. A momentum-based introspective step then encourages these perturbed features to stay close to a more stable, generalized embedding. The key takeaway is practical: instead of trying to handcraft the exact weather corruption in input space, DGUIL trains robustness directly against a family of feature-level variations.

NormFusion [67] addresses the same gap at deployment time by adapting **feature statistics** rather than features themselves. In streaming TTA, it fuses batch-normalization statistics from training and from the current test sample in a stage-wise manner: earlier layers lean more on test-time statistics to follow the shifted distribution, while later layers retain training statistics to avoid drifting high-level semantics. The takeaway is that a careful normalization strategy can recover performance under weather shift with minimal overhead, without changing the raw input. Such normalization-based TTA is attractive for real-time deployment because it offers lightweight online correction under weather shift. However, it mainly addresses feature-statistics drift. When adverse weather causes severe missing returns, clutter, or unstable semantic evidence, statistical calibration alone may not be enough to fully stabilize prediction.

Overall, feature-space modeling offers a complementary handle to input-level weather simulation. It either teaches the network to tolerate uncertainty by training on perturbed internal features, or it reduces the clear-to-adverse gap by calibrating feature statistics at test time. In both cases, the goal is to keep the representation well-conditioned under shift, so that downstream decoding does not amplify local corruption into widespread prediction errors.

5.2.3. Semantic Structure Alignment in Feature Space

Even when feature extraction is made more stable under adverse weather, models can still fail because the *semantic geometry* of the feature space collapses: classes start to overlap, decision boundaries blur, and rare or safety-critical things are easily absorbed into dominant stuff regions. Methods in this group address this failure mode directly by **reshaping and preserving class structure in feature space**—pulling same-class features closer across conditions and pushing different classes apart—so that adaptation does not come at the cost of discriminability. In practice, this is often implemented with explicit anchors such as prototypes or memory banks, feature binding or distillation that ties corrupted features back to clean structure, and overlap-aware alignment mechanisms that discourage cross-class mixing. The common goal is simple: keep the representation space well-separated and semantically consistent, so weather corruption does not turn into systematic misclassification.

Building on the strong weather augmentation baseline SJ+LPD [38], NTN [39] shifts the focus from "making the input harder" to **keeping the semantic structure of features intact**, especially for safety-critical **things** classes. The authors point out that under adverse weather, things are often misclassified as visually dominant stuff, because corruption degrades both semantic-level features and local evidence. To counter this, NTN introduces two simple feature-space constraints. First, **Feature Binding** groups visually related classes into superclasses and pulls corrupted point features toward superclass prototypes, which reduces catastrophic things→stuff confusion. Second, **Beam-wise Feature Distillation** uses each LiDAR beam as a local region and aligns features from clean and corrupted counterparts at the beam level, providing stronger learning signals when points are missing or locally corrupted.

A natural next step is to ask what happens when **alignment itself** starts to blur class boundaries. Chen et al. [4] makes this point explicit: in unsupervised adaptation, shared classes can drift into each other because boundaries are not well aligned, and overly "domain-invariant" training can wash out target-specific structure. To address this, the paper adds an explicit anti-overlap mechanism on top of teacher–student UDA: an invertible attention alignment module that performs bidirectional mapping while suppressing cross-class overlap, and a quality-guided contrastive memory bank that filters unreliable features and sharpens class-wise prototypes. The message is clear and useful for adverse weather: it is not enough to pull features together—you also need a guardrail that keeps different classes separated when the target domain is noisy.

If overlap is the symptom, a complementary way to address it is to make the feature extractor less easily distracted by corrupted evidence. Liao et al. [32] inserts a Point Attention Block into a PointDR-style pipeline, so the network can reweight local neighborhoods and focus on the cues that remain informative when weather distorts point patterns. In a different setting, He et al. [21] brings the same spirit to cross-weather generalization by learning **category-level** geometric embeddings and enforcing geometric-consistent alignment between original and weather-simulated scans, so that class-wise prototypes remain stable and the feature space preserves semantic structure under adverse-weather shifts. The shared takeaway is that preserving class separation is not only about explicit anti-overlap losses—it also benefits from either more selective feature extraction (attention) or stronger geometric structure during adaptation.

Overall, semantic structure alignment methods treat adverse weather as a threat to **class separability** rather than only to feature stability. Whether through feature binding and distillation, explicit anti-overlap alignment with prototype memories, or more selective and structure-aware adaptation, they share the same objective: keep the feature space well organized so that "things" are not swallowed by "stuff" and boundaries do not collapse under corruption. This perspective is especially valuable in adverse weather, where the input evidence is uneven and noisy, and naive alignment can easily trade accuracy for confusion.

5.2.4. Summary of B2

B2 methods improve adverse-weather robustness by shaping the **representation space** rather than the raw input. Their main advantage is that they can explicitly handle cue asymmetry and observation unreliability: decouple geometry/radiometry to limit cross-cue contamination, calibrate features or statistics to reduce drift, and preserve class structure to prevent boundary collapse under corruption. As a result, B2 often complements B1 by improving robustness even when the input corruption pattern is not perfectly modeled.

The limitations are that B2 designs can be sensitive to supervision quality and domain noise: unreliable intensity cues, biased normalization statistics, or noisy correspondences/prototypes may make fusion or alignment error-amplifying. In addition, extra branches, memories, or calibration modules add complexity and may require careful tuning across sensors and weather regimes.

5.3. Weather-Robust Objective and Prediction Constraints (B3)

B3 methods improve adverse-weather robustness by intervening in the **objective / prediction space**, i.e., by redesigning the training signal, loss functions, or prediction-level constraints that govern how the model is optimized. Compared with B1 input preprocessing and B2 representation learning, B3 does not primarily alter the observation or the feature extractor; instead, it explicitly encodes **what predictions should remain stable** under weather-induced corruption and **which predictions should be trusted** when supervision is scarce or unreliable. This perspective is well aligned with adverse-weather LiDAR shift: weather often removes or corrupts evidence (e.g., clutter, truncation, wet-ground holes), so robustness hinges on controlling confirmation bias from noisy pseudo labels and preventing over-confident drift during adaptation. In this survey, we divide B3 methods into two complementary families based on a simple criterion: whether they create an explicit pseudo-supervision loop on target-domain data. **Pseudo-supervision / self-training** methods turn unlabeled (or weakly labeled) adverse-weather scans into training signal by generating pseudo labels and updating the segmenter with reliability control (e.g., confidence/purity filtering, temporal smoothing, and robust losses), which naturally fits UDA and TTA protocols. In contrast, **constraint- and risk-aware regularization** methods do not rely on pseudo labels as supervision; instead, they directly shape the optimization objective by enforcing prediction invariances (e.g., consistency across density views or temporal correspondences) and/or by reweighting the loss to emphasize structurally fragile regions, making them effective even in source-only robustness settings. In the remainder of this section, we review representative methods under these two directions and discuss their respective trade-offs. Table 4 summarizes representative B3 papers by learning setting and the specific objective / prediction operations they apply.

5.3.1. Pseudo-supervision / Self-training on Adverse-Weather Data

Methods in this group improve adverse-weather robustness by turning **predictions into supervision**. The pipeline is characterized by an explicit pseudo-supervision loop: the model (or an EMA teacher) first produces pseudo labels on unlabeled (or weakly labeled) adverse-weather scans, then updates the segmenter using only the subset of predictions deemed reliable. This design is particularly natural under adverse weather, where the dominant failure mode is not a lack of representation capacity but a lack of trustworthy evidence: near-range clutter, far-range truncation, and wet-ground dropouts can make naive self-training collapse into confirmation bias. As a result, successful pseudo-supervision methods spend most of their effort on **reliability control** rather than on the pseudo-labeling act itself—for example, combining confidence with local geometric agreement, stabilizing updates with temporal smoothing on streams[68], and using robust objectives or warmup schedules so that early, noisy pseudo labels do not dominate optimization[41]. Within this family, existing methods differ less in **whether** to use pseudo labels and more in **how** to make them reliable under adverse weather.

HGL [68] tackles pseudo-label reliability in streaming TTA by anchoring it to hierarchical geometry. It first forms **local** pseudo labels by aggregating predictions within k-NN neighborhoods and scores each region by combining prediction certainty with **geometric purity** (neighbor agreement). It then refines these labels at a **global** level using class prototypes updated on the target stream (EMA), keeping only labels that agree between the local and prototype assignments. To stabilize online updates, HGL further adds a temporal regularizer that aligns geometrically matched points across frames and biases the alignment so that low-confidence features are pulled toward high-confidence ones.

HGL also illustrates why TTA is practically attractive for adverse-weather deployment: it can adapt on the incoming target stream without requiring target labels or offline retraining. At the same time, it highlights a central difficulty of weather-specific TTA, which remains relatively underexplored compared with DG and UDA. The main challenge is not only how to design the update rule, but how to ensure that the self-supervised adaptation signal remains reliable when adverse weather severely corrupts the input. Once pseudo labels or temporal correspondences become unstable, online adaptation can accumulate errors and drift.

While HGL assumes no target labels, Piroli et al. [41] address a common field setting: abundant adverse-weather scans but only a handful of annotations. They start from few-shot supervision on K labeled target frames, then switch to a semi-supervised loop that generates pseudo labels for the remaining unlabeled data and trains with an SSL objective built to tolerate noisy pseudo labels (symmetric cross entropy), together with a warmup schedule that delays strong reliance on pseudo supervision. To make training less brittle under tiny labeled sets, they further use pseudo-validation splits for checkpoint selection in both stages. SSMix [11] adopts a teacher–student scheme and uses consistency to make pseudo supervision usable. An EMA teacher predicts pseudo labels from weakly augmented target scans; only predictions above the confidence threshold are kept, and the student is trained to match them under strong augmentation via a consistency loss.

The core challenge for this family is not how to **produce** pseudo labels, but how to make them **trustworthy** under adverse weather. As weather corrupts and removes evidence, naive self-training easily amplifies early mistakes. Effective

Table 4
B3 (Objective / Prediction-space) methods for adverse-weather LiDAR semantic segmentation.

Setting	Reference	Year	Weather focus	Objective / prediction operation
DG	Robo3D [29]	2023	fog/rain/snow/wet ground	Density-insensitive training objective: cross-density completion and confirmation losses between full and partial views
TTA	HGL [68]	2024	fog/snow	Local–global pseudo-label generation and filtering + prototype-based pseudo-label refinement + temporal consistency regularization
UDA	Piroli et al. [41]	2024	fog/snow/spray	Pseudo-label self-training with robustness to noisy pseudo labels + pseudo-validation model selection
UDA	SSMix [11]	2024	rain/fog/snow	Teacher–student (EMA) pseudo-labeling with confidence thresholding + weak/strong augmentation consistency loss integrated into total objective
DG	Cheong et al. [6]	2025	rain/fog/snow	Geometry-aware risk reweighting after point drop + entropy regularization to discourage over-confidence

pseudo-supervision therefore centers on reliability control, filtering pseudo labels by confidence and geometric agreement, stabilizing them over time.

5.3.2. Constraint- and Risk-aware Regularization under Adverse Weather

Methods in this group improve adverse-weather robustness by **shaping the objective directly**, without relying on an explicit pseudo-label supervision loop. The common premise is that adverse weather changes the evidence far more than the underlying scene semantics: points may disappear, boundaries break, and local neighborhoods become unreliable. Rather than treating the model's predictions as supervision, these methods inject **predictive constraints** and **risk-aware weighting** into training so that optimization is steered toward invariances that should hold under weather-induced degradation. Compared with pseudo-supervision, this family is typically easier to deploy in source-only settings and less prone to confirmation bias; the main challenge is to design constraints and risk signals that remain informative when weather is severe.

Robo3D [29] views adverse-weather degradation largely as **structured point loss**, and therefore targets invariance to density changes rather than crafting weather-specific heuristics. It trains on paired full and randomly masked partial scans in a teacher–student setup, and augments the per-view segmentation loss with cross-density consistency. Specifically, a **completion** term aligns the partial prediction to the full prediction, while a **confirmation** term enforces local agreement at the partial resolution. This objective directly suppresses prediction drift caused by density drop, without relying on a pseudo-label self-training loop.

Cheong et al. [6] take a risk-driven view: adverse weather rarely corrupts the scan uniformly, but first disrupts *structurally fragile* regions such as boundaries, corners, and already-sparse areas, where small mistakes quickly cascade into semantic confusion. They encode this insight directly in the objective by deriving a per-point vulnerability score from lightweight geometric cues (local-window KNN with circular wrapping for neighborhood continuity) and using it to reweight the post-drop segmentation loss, up-weighting high-risk points. An entropy regularizer is added to reduce over-confident errors.

The core idea of this family is to make robustness come from the **objective itself**, rather than from an explicit pseudo-label loop. The takeaway is that, when target supervision is unavailable or unreliable, carefully designed constraints and risk-aware loss shaping can provide a stable robustness signal while avoiding confirmation bias.

5.3.3. Summary of B3

B3 methods target robustness through the **training signal** rather than the input or backbone. Their main advantage is that they directly control what the model should trust and what should stay stable under corruption: pseudo-supervision turns target predictions into usable labels with filtering/smoothing, while constraint- and risk-aware objectives provide supervision-free regularization that is less prone to confirmation bias and can work in source-only settings.

The limitations are that pseudo-supervision can still be brittle when weather is severe—noisy pseudo labels and over-confident drift may accumulate despite filtering. Constraint-based objectives are usually safer but may underperform if the chosen invariances or risk signals are weak proxies for real weather failure modes, and they often require careful tuning to balance stability and plasticity.

Compared with DG and UDA, TTA remains relatively underexplored in adverse-weather LiDAR semantic segmentation, despite its strong practical relevance. This is important because real deployments rarely face a fixed target domain; instead, weather type and severity may evolve continuously over time, making online adaptation especially attractive. Existing TTA methods show that lightweight correction through normalization calibration or streaming self-supervision can improve robustness, but they also reveal the central difficulty of this setting: once the self-supervised signal becomes unreliable under severe corruption, online updates may accumulate errors and drift. As a result, effective TTA under adverse weather must balance plasticity,

Table 5
Cross-domain performance (mIoU, %) on *SemanticSTF*. All results are reported under the **standard 19-class evaluation protocol** (moving classes merged; rare/invalid labels ignored) as used in the referenced papers, and all methods use the MinkUNet/MinkNet backbone.

Setting	Paper	Tag	SemanticKITTI	SemanticKITTI→SemanticSTF	SynLiDAR→SemanticSTF
DG	PointDR [57]	B1	–	28.4	18.5
DG	DGUIL [20]	B2	–	35.5	21.1
DG	UniMix [65]	B1	61.5	31.4	23.4
DG	GRC [61]	B2	63.1	42.5	23.5
DG	NTN [39]	B2	–	38.9	–
DG	Park et al. [38]	B1	62.8	39.5	20.1
DG	Cheong et al. [6]	B3	–	39.2	–
DG	Liao et al. [32]	B2	–	30.0	–
DG	He et al. [21]	B2	–	37.5	26.3
UDA	UniMix [65]	B1	61.5	37.2	26.7
UDA	SSMix [11]	B3	–	32.3	20.0

stability, efficiency, and safety more carefully than offline adaptation.

6. Benchmarking Adverse-Weather Generalization: Results and Future Directions

While Sections 3–5 categorized methods by assumptions and intervention levels (B1–B3), a complementary question remains: **how much robustness do these design choices translate into under standardized cross-domain tests?** We therefore summarize reported mIoU on *SemanticSTF*, a real-world adverse-weather benchmark (fog/rain/snow) that is most commonly used in recent studies. To improve comparability, we focus on results that largely follow the evaluation protocol established by PointDR[57] and adopted by most subsequent works in this area. In particular, for SemanticKITTI→SemanticSTF, we follow the widely used **19-class protocol** (merging moving labels, ignoring rare/undefined categories, and excluding *invalid* points), and we restrict entries to methods reported with the **MinkUNet/MinkNet backbone family**. Most of the included papers also follow the same practical training and testing setup, namely training on SemanticKITTI sequences 00–07 and 09–10 and evaluating on SemanticSTF. For SynLiDAR→SemanticSTF, the reported results are also largely based on the commonly used *subset* version of SynLiDAR (approximately 20,000 frames), rather than the full dataset. The numbers in Table 5 are collected from the original papers rather than reproduced under a unified codebase. Therefore, although the compared methods are not arbitrarily selected and already share a largely consistent protocol, some differences in preprocessing, augmentation, optimization, and model selection may still remain across studies. The table should thus be read as a comparison under substantially aligned settings, with the goal of highlighting broad empirical trends rather than claiming a perfectly controlled one-to-one ranking. We then present performance trends by learning setting (DG vs. UDA) and distill practical takeaways and open challenges.

6.1. Cross-domain Benchmarking on SemanticSTF

This subsection provides a quantitative comparison of representative adverse-weather adaptation methods under a unified cross-domain evaluation protocol. As discussed in Sec. 4.2, the comparison is primarily based on mean Intersection-over-Union (mIoU), which reflects the overall agreement between predicted semantics and ground truth across the evaluated classes. Table 5 summarizes the reported mean IoU (mIoU) on *SemanticSTF* under two widely used transfer settings: **SemanticKITTI→ Semantic STF** and **SynLiDAR→SemanticSTF**. Crucially, we also include the performance on the source domain (labeled as 'SemanticKITTI') to evaluate the stability of these methods, ensuring that robustness improvements do not come at the cost of degrading performance on standard clear-weather data. All results are reported on the **19-class** label set and are **stratified by learning setting** (**DG** vs. **UDA**), since these protocols differ in whether target-domain scans are accessible during training and are therefore not directly interchangeable. To improve comparability, we only include entries evaluated with the **MinkUNet/MinkNet backbone** as reported in the corresponding papers. For each method, we further annotate its **primary intervention tag** (B1–B3) to link the numerical outcomes back to the taxonomy in Sec. 5. A dash ("–") indicates that the corresponding transfer result is not reported and is thus excluded from comparison. The purpose of Table 5 is not to provide an absolute ranking, but to surface trends and their implications for safety-critical classes and operational reliability.

Overall, Table 5 shows a clear separation between the two transfer settings. On **SemanticKITTI→SemanticSTF**, the reported mIoU spans a wide range from **28.4** to **42.5**, with the best result achieved by **GRC** [61] (42.5) and several recent methods forming a strong second tier around **37.5–39.5**. On the more challenging **SynLiDAR→SemanticSTF**

transfer, performance is uniformly lower, ranging from **18.5** to **26.7**. The strongest results are obtained by **UniMix** [65] under UDA (26.7) and **He et al.** [21] under DG (26.3), while most other reported numbers concentrate in the **20–24** band. This gap is consistent with the fact that SynLiDAR→Semantic STF compounds synthetic-to-real discrepancies with adverse-weather sensing degradations, whereas SemanticKITTI→ SemanticSTF isolates weather shift more directly.

Taken together, Table 5 highlights different empirical behaviors under DG and UDA. Under **DG**, several methods trained solely on labeled clear-weather sources already transfer well to real adverse-weather scans: on SemanticKITTI→SemanticSTF, the best result reaches **42.5** mIoU (GRC), and multiple approaches cluster in the high-**30**s (**37.5–39.5**), suggesting that a large portion of the weather gap can be mitigated without target access when the method explicitly accounts for measurement unreliability. In contrast, input-level randomization baselines such as PointDR remain substantially lower (28.4), indicating that broad perturbations may not sufficiently cover the structured degradations observed in real adverse weather. Under **UDA**, where unlabeled target scans are available, the benefit is more apparent in the harder SynLiDAR→SemanticSTF transfer: UniMix under UDA achieves the strongest reported mIoU (**26.7**), exceeding most DG counterparts and underscoring the value of exploiting target observations when the shift compounds sim-to-real discrepancies with adverse-weather degradation. However, for SemanticKITTI→Semantic STF, the reported UDA numbers (**37.2** for UniMix and **32.3** for SSMix) do not exceed the strongest DG results, which is consistent with the difficulty of pseudo-supervision under adverse weather: clutter, missing returns, and radiometric distortions can make pseudo labels brittle, so additional target data does not automatically translate into higher mIoU without stringent reliability control.

Looking across the intervention tags, Table 5 suggests a consistent mechanism-level pattern. **B2** approaches are generally the most effective, indicating that robustness under adverse weather is often won in the *feature space*: explicitly decoupling heterogeneous cues, calibrating feature statistics, or maintaining discriminative class structure helps prevent early corruption from propagating through the network and stabilizes downstream decoding. **B1** methods offer a strong and portable baseline by expanding the input distribution with weather-aware perturbations or mixing, but their improvement tends to saturate when degradations are dominated by missing or unreliable evidence rather than removable noise. **B3** objectives provide an orthogonal lever by enforcing invariances or leveraging pseudo supervision, yet their success depends critically on reliability control, since adverse weather can make pseudo labels and consistency targets brittle. Overall, the table points to a practical takeaway: input-level coverage matters, but the most reliable gains typically come from representation- and objective-level mechanisms that explicitly manage uncertainty and prevent error amplification under structured weather corruption.

Beyond the aggregate numbers in Table 5, we provide a qualitative comparison in Fig. 7. To avoid possible misunderstanding, Fig. 7 is intended as an illustrative qualitative example based on representative methods, rather than an exhaustive visual comparison across all intervention categories. The example highlights a common pattern on SemanticSTF: adverse weather produces incomplete and uneven evidence, so errors tend to concentrate around ambiguous regions and long-range structures. Compared with the baseline PointDR(B1), DGUIL(B2) yields cleaner and more coherent predictions in the highlighted regions, with fewer spurious fragmented labels and better preservation of local semantic structure. This qualitative gap is consistent with the mechanism-level trend observed in Table 5: methods that explicitly shape intermediate representations (B2) often translate more reliably to adverse-weather data than input-only perturbation baselines (B1), especially when the dominant difficulty is missing or unreliable returns.

6.2. Summary of Findings and Future Research Directions

Adverse weather induces a *structured* shift in LiDAR observations rather than simple, i.i.d. noise: clutter, truncation, wet-surface dropouts, and radiometric distortions jointly change what evidence is available and how reliable that evidence is. Across the literature reviewed in Secs. 3–5, a recurring conclusion is that robust segmentation under such shifts is rarely achieved by a single lever. Input-space strategies (B1) are valuable for expanding coverage of weather-like degradations, but they often plateau when the dominant failure mode is missing or ambiguous evidence. Conversely, methods that explicitly manage reliability at the representation and objective levels (B2/B3) tend to deliver more consistent gains by stabilizing feature extraction and constraining optimization against error amplification. In practice, the most dependable recipe is increasingly hybrid: combine weather-aware input exposure with feature- or loss-level mechanisms that gate unreliable cues and prevent confirmation bias, rather than relying solely on heavier backbones or aggressive self-training.

6.2.1. Direction 1: Beyond Handcrafted Weather Simulation Toward Statistical Feature-space Canonicalization

A direct implication of the above synthesis is that adverse-weather robustness should not depend mainly on handcrafted B1 corruptions, because real weather is highly variable and difficult to parameterize in a way that remains faithful across scenes, sensors, and severities. A more scalable alternative is to **reduce domain bias in representation space through statistical canonicalization**. The key idea is to treat adverse weather as a shift in the **distribution** of intermediate features and to explicitly map features to a more stable, domain-agnostic form using lightweight, statistically grounded operations. Concretely, one can (i) normalize and align feature statistics at appropriate stages (beyond vanilla BN) using instance-/group-wise statistics,

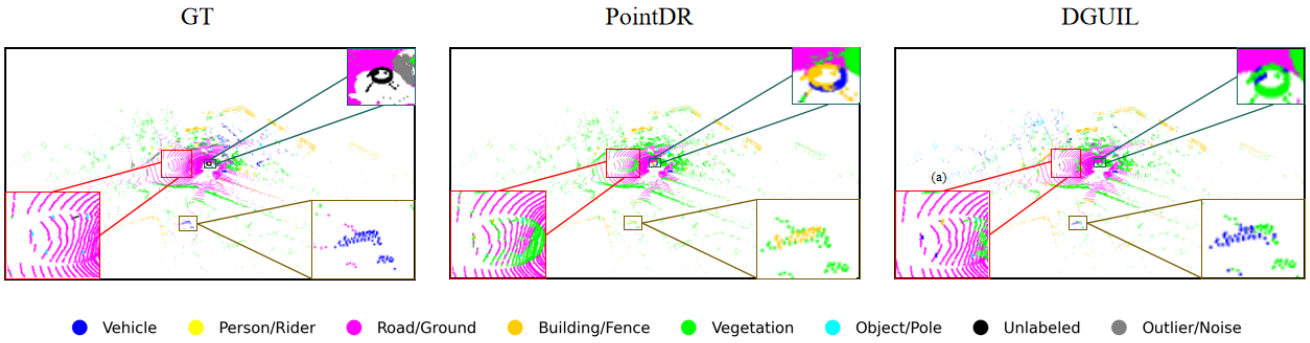


Figure 7: Qualitative comparison on SemanticSTF under adverse weather. From left to right: ground truth (GT), PointDR[57], and DGUIL[20]. Zoomed-in regions illustrate typical adverse-weather failure modes and the relative differences between methods.

robust moments, or running estimates; (ii) apply stochastic feature perturbation driven by learned feature uncertainty so that the backbone is trained against a family of plausible feature distributions rather than a single "clean" one; and (iii) regularize intermediate features toward a canonical distribution while preserving class separability, so that weather-induced bias is absorbed early and does not propagate to the decoder. This direction targets the **mechanism** of domain shift (feature distribution drift) rather than its unpredictable surface manifestation in the input.

6.2.2. Direction 2: Physics-consistent Radiometry Modeling and Decoupled Intensity Usage

A second direction follows directly from the measurement analysis in Sec. 3.3: adverse weather induces large and condition-dependent **radiometric bias** that makes the intensity channel far less stable than geometry. Extinction, backscattering, wet-surface effects, and device-dependent gain control can all reshape intensity statistics, so treating intensity as a generic, stationary input feature is an avoidable source of brittleness. Future progress therefore likely requires **physics-consistent** handling of radiometry, coupled with **explicit decoupling** between geometric and intensity representations. Concretely, intensity should be modeled and used conditionally—e.g., via separate encoding streams, calibration modules that correct distribution shifts, and reliability-gated fusion mechanisms that regulate when and where intensity cues are trusted. Such designs align with the fact that weather primarily changes the sensing process rather than the scene semantics, and they offer a principled way to benefit from intensity when it is informative while limiting its harm under severe radiometric distortion.

6.2.3. Direction 3: Safe and Efficient Test-Time Adaptation under Non-stationary Weather

A third direction is to develop **safe and efficient test-time adaptation** specifically for adverse-weather LiDAR segmentation in real driving. Unlike offline adaptation, real deployment does not face a fixed target domain. Instead, weather conditions may change continuously within a single trip, so the model must adapt under a **non-stationary** target stream rather than a predefined dataset. This makes TTA particularly attractive in practice, since it offers a way to correct distribution shift online without requiring target labels or costly retraining.

At the same time, adverse-weather TTA is difficult to make reliable. Under severe clutter, range truncation, or unstable semantic evidence, pseudo labels, temporal correspondences, or other self-supervised cues may become unreliable, which can lead to error accumulation and model drift. Future progress therefore likely depends on TTA designs that are not only adaptive, but also **confidence-aware, parameter-efficient, and deployment-compatible**. Promising directions include lightweight updates on a small subset of parameters or feature statistics, confidence-gated adaptation, temporal-consistency mechanisms with drift prevention, and reset or recovery strategies when online adaptation becomes unstable.

7. Conclusion

This survey reviewed adverse-weather LiDAR semantic segmentation through a unified framework that connects learning settings (DG/UDA/TTA), physics-driven domain shifts, and pipeline intervention levels (B1–B3). The central finding is that adverse weather induces structured degradations—clutter, range truncation, wet-surface dropouts, and radiometric distortion—so robustness cannot be achieved by generic noise tolerance alone: input-level exposure (B1) is helpful but often saturates under missing or unreliable evidence, whereas representation- and objective-level mechanisms (B2/B3) more consistently mitigate error amplification, as also reflected on SemanticSTF. Looking ahead, promising directions include statistically grounded feature-space canonicalization and physics-consistent, reliability-gated use of radiometry to prevent intensity bias from propagating through the network.

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